CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

May 2014

CL 210 Hydrology & Hydraulics

Date: 07.05.2014, Wednesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain the following terms: [03]

(i) Evaporation (ii) Transpiration (iii) Evapotranspiration

(b) The average rainfall over 45 ha. of watershed for a particular storm was as follows: [04]

Time from start (hr)	1	2	3	4	5	6	7
Incremental rainfall in each hour (cm)	0.5	1.0	3.25	2.5	1.5	0.5	0.0

Total runoff volume from this storm was determined as 2.25 ha.m. Determine Φ index.

Q – 2(a) A 30 cm well fully penetrates a confined aquifer 30 m deep. After a long period of pumping at a rate of 1200 lpm, the drawdown in the wells at 20 and 45m from the pumping well are found to be 2.2 and 1.8 m, respectively. Determine the transmissibility of the aquifer. What is the drawdown in the pumped well? [07]

(b) Enlist various sub-surface techniques of ground water investigation. Discuss any two techniques in detail. [07]

OR

Q – 2(a) Enlist various surface techniques of ground water investigation. [03]

(b) Explain the following terms: [04]

(i) Aquitard (ii) Aquifuge (iii) Aquiclude (iv) Aquifer

(c) Explain unsteady radial flow in to a well situated in an unconfined aquifer. [07]

Q – 3(a) What is salt water intrusion? How it is caused? [03]

(b) Explain the concept of 'Upconing of saline water'. [04]

(c) Enlist various artificial methods of groundwater recharge. Explain, in detail, any [07]

TWO of them.

OR

Q – 3(a) What are the needs of ground water recharge? [03]

(b) Discuss briefly Chemical Qualities of ground water. [04]

(c) A well screen 1 m in length is located 10 m below the ground water table in an unconfined aquifer having a permeability of 22.5 m/day. The fresh water-sea water interface exists at a depth of 40 m below the water table. What is the maximum discharge that can be sustained from the well without causing the salt water to intrude into the well? [07]

SECTION – II

Q – 4(a) What are the disadvantages of Roof Rain Water Harvesting System? [03]

(b) Define: Conjunctive Use. What are the needs of conjunctive use? [04]

Q - 5 (a) Enlist the laboratory methods of measurement of permeability. [03]

(b) Which are the factors affecting ground water pollution? [03]

(c) Which are the water quality criteria for domestic use as per IS and WHO standards? [03]

(d) State and derive the Darcy's Law for movement of ground water. [05]

OR

Q - 5 (a) Explain the following terms: [03]

(i) Transmissibility (ii) Storage Coefficient (iii) Porosity

(b) In a certain Place of Madhya Pradesh, the average thickness of the aquifer of the confined aquifer is 20 m and extends over an area of 720 km^2. The piezometric surface fluctuates annually from 22 to 15 m above the top of aquifer. What ground water can be expected annually, if a storage coefficient is 0.00072? Assuming, an average well yield of 25 m^3/hr and about 150 days of pumping in a year, how many wells can be drilled in the area? [05]

(c) Which parameters and limits have been recommended to judge the quality of water used in irrigation? Discuss in detail. [06]

Q - 6 (a) Classify rain gauges used for measurement of rainfall. [03]

(b) Enlist various factors affecting runoff. [03]

(c) Explain the components of Roof Rain Water Harvesting System with neat sketch in detail. [03]

(d) The following are the rates of rainfall for successive 20 minutes period of 140 minutes storm: 2.5, 2.5, 10.0, 7.5, 1.25, 1.25, 5.0 cm/hr. Taking the value of Φ index [05] as 3.2 cm/hr. Find out the net runoff and total rainfall in cm.

OR

Q - 6 (a) Draw a neat sketch of Symon's rain gage. [03]

(b) Enlist methods of flood forecasting. [03]

(b) Enlist the techniques for conservation of surface runoff. [03]

(c) Given the ordinates of a 4-hr unit hydrograph as below. Derive the ordinates of a 12-hr unit hydrograph for the same catchment by S-curve method. [05]

Time (h)	0	4	8	12	16	20	24	28	32
Ordinate of 4-h UH (m^3/s)	0	20	80	130	150	130	90	52	27
Time (h)	36	40	44						
Ordinate of 4-h UH (m^3/s)	15	5	0						

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